

**Question 1.****Marks**

- (a) Find (i)  $\int \sin(ax + \frac{\pi}{6}) dx$ . 1
- (ii)  $\int \sin^2(ax + \frac{\pi}{6}) dx$ , where  $a$  is a constant. 2
- (b) Find  $\int \frac{dx}{\sqrt{x^2 - 6x + 13}}$ . 3
- (c) Evaluate  $\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \sin^{-1}(x) dx$ . 3
- (d) (i) Given that  $\frac{2x^2 - 2x + 1}{(x+1)(2x^2 + 3)} = \frac{1}{x+1} + \frac{ax+b}{2x^2 + 3}$ , 2
- find the real numbers  $a$  and  $b$ .
- (ii) Hence, or otherwise evaluate  $\int_0^{\frac{\sqrt{3}}{\sqrt{2}}} \frac{2x^2 - 2x + 1}{(x+1)(2x^2 + 3)} dx$ . 4

**Question 2. [START A NEW PAGE]**

- (a) Given  $f(x) = x^2 - 2x$ , sketch the graph of the following on separate axes
- (i)  $y = f(x)$ . 1
- (ii)  $y = f(x) + |f(x)|$ . 2
- (iii)  $y^2 = f(x)$ . 2
- (iv)  $y = [f(x)]^{-1}$ . 2
- (v)  $y = f(2^x)$ . 2
- (b) (i) For the Hyperbola:  $H: \frac{x^2}{9} - \frac{y^2}{16} = 1$ , find its eccentricity  $e$ . 2
- (ii) Hence, neatly sketch the graph of  $H$ , clearly showing the vertices, foci, directrices and asymptotes. 4

**Question 3.****[START A NEW PAGE]****Marks**

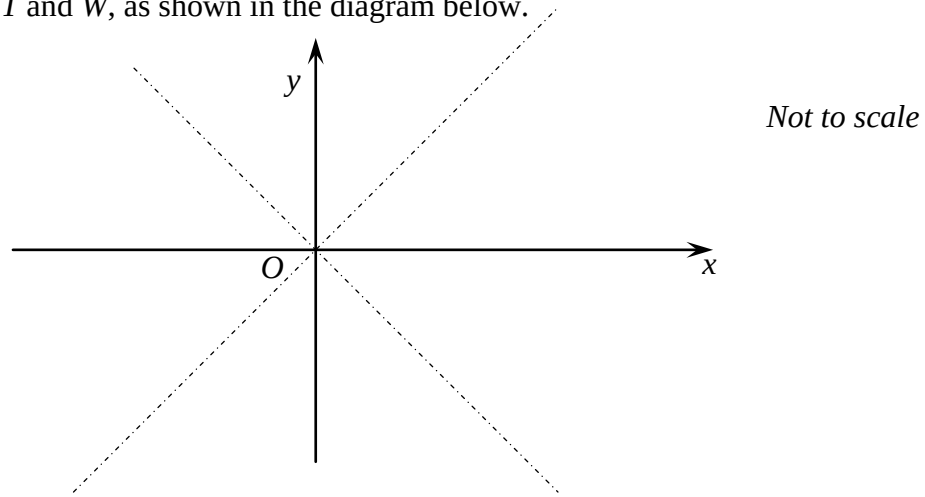
- (a) An ellipse  $E$  has the equation:  $E: \frac{x^2}{100} + \frac{y^2}{75} = 1$ .
- (i) Sketch this ellipse, clearly showing on your diagram the coordinates of the foci and the equation of each directrix. **4**
- (ii) Show that the equation of the normal to the ellipse  $E$  is  $4x - 2y = 5$  at point  $P(5, 7\frac{1}{2})$ . **2**
- (iii) A circle is tangential to the ellipse  $E$  at  $P$  and at  $Q(5, -7\frac{1}{2})$ . **1**  
Show that the centre of the circle is  $(1\frac{1}{4}, 0)$ .
- (iv) Hence find the equation of this circle. **2**
- (b) Using the substitution:  $t = \tan \frac{x}{2}$ , where  $0 \leq x < \pi$ , show that: **4**
- $$\int_0^{\frac{2\pi}{3}} \frac{dx}{5 + 3 \sin x + 4 \cos x} = \frac{\sqrt{3} - 1}{3}.$$
- (c) For  $x > 0$ , find:  $\int \frac{dx}{x^4 + x}$ . **2**

**Question 4.**

**[START A NEW PAGE]**

**Marks**

- (a) The point  $S(3e, 0)$  is a focus of the hyperbola  $H: x^2 - y^2 = 9$ .  
The tangent to the hyperbola, at the point  $P(p, q)$ , meets the asymptotes of  $H$  in  $T$  and  $W$ , as shown in the diagram below.



- (i) Show that the equation of the tangent  $TW$  is given by:  $px - qy = 9$ . **1**
- (ii) Show that the gradient of the line through  $SW$  is given by **1**
- $$m_{SW} = \frac{3}{e(p+q)-3}.$$
- (iii) By letting  $\angle WST = \theta$ , find the value of  $\tan \theta$ . **3**

- (b) Let  $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$ , for  $n = 0, 1, 2, \dots$

- (i) Show that  $I_1 = \frac{1}{2} \ln 2$ . **2**
- (ii) Show that,  $I_n + I_{n-2} = \frac{1}{n-1}$ , for  $n = 2, 3, 4, \dots$  **3**
- (iii) Explain why  $I_n < I_{n-2}$ , for  $n \geq 2$ , and deduce that **3**

$$\frac{1}{2n+2} < I_n < \frac{1}{2n-2}.$$

- (iv) By using the recurrence relation in part (b) (ii), **2**  
find  $I_5$ , and deduce that:  $\frac{2}{3} < \ln 2 < \frac{3}{4}$ .

**THE END**



# Question 1

$$i) \int \sin(ax + \frac{\pi}{6}) dx = -\frac{1}{a} \cos(ax + \frac{\pi}{6}) + C \quad (1)$$

$$ii) \int \sin^2(ax + \frac{\pi}{6}) dx = \frac{1}{2} \int [1 - \cos 2(ax + \frac{\pi}{6})] dx \quad (1)$$

$$= \frac{x}{2} - \frac{1}{4a} \sin 2(ax + \frac{\pi}{6}) + C \quad (1)$$

$$iii) \int \frac{1}{x^2 - 6x + 13} dx$$

$$= \int \frac{1}{(x-3)^2 + 4} dx \quad (1)$$

$$= \frac{1}{2} \tan^{-1}(\frac{x-3}{2}) + C \quad (1)$$

$$\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \sin^{-1} x dx = [x \sin^{-1} x]_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} - \int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{x}{\sqrt{1-x^2}} dx$$

$$= \left[ \frac{\sqrt{3}}{2} \sin^{-1} \frac{\sqrt{3}}{2} - \frac{1}{2} \sin^{-1} \frac{1}{2} \right] + \int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{d(1-x)}{\sqrt{1-x^2}}$$

$$= \left[ \frac{\sqrt{3}\pi}{6} - \frac{\pi}{12} + \frac{1}{2} [(1-x^2)^{-\frac{1}{2}}]_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \right]$$

$$= \frac{2\sqrt{3}\pi - \pi + 3 - 3\sqrt{3}}{12}$$

$$iv) \frac{2x^2 - 2x + 1}{(x+1)(2x^2+3)} = \frac{1}{x+1} + \frac{ax+b}{2x^2+3}$$

$$RHS = \frac{2x^2+3+(ax+b)(x+1)}{(x+1)(2x^2+3)} \quad (1)$$

$$= \frac{(2+a)x^2 + x(a+b) + b+3}{(x+1)(2x^2+3)}$$

equating coefficients:

$$2 = 2+a$$

$$-2 = a+b$$

$$1 = b+3$$

$$\therefore a = 0$$

$$\text{and } b = -2 \quad (1)$$

$$v) \int_0^{\frac{\sqrt{3}}{2}} \left[ \frac{1}{x+1} + \frac{-2}{2x^2+3} \right] dx \quad (1)$$

$$= \left[ \ln(x+1) + \frac{-1}{\sqrt{3}} \tan^{-1} \left( \frac{\sqrt{3}}{2} x \right) \right]_0^{\frac{\sqrt{3}}{2}}$$

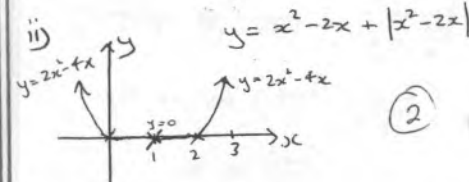
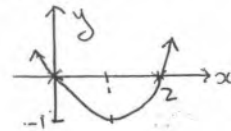
$$= \left[ \ln(x+1) \right]_0^{\frac{\sqrt{3}}{2}} + \frac{-\sqrt{3}}{3} \left[ \tan^{-1} \frac{\sqrt{3}x}{\sqrt{3}} \right]_0^{\frac{\sqrt{3}}{2}} \quad (1)$$

$$= \ln \left( \frac{\sqrt{3}}{2} + 1 \right) + \frac{-\sqrt{3}}{3} (\tan^{-1} 1 - \tan^{-1} 0) \quad (1)$$

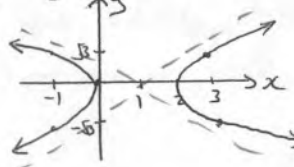
$$= \ln \left( \frac{\sqrt{3}}{2} + 1 \right) + \frac{-\sqrt{3}}{3} \frac{\pi}{4} \quad (1)$$

## Question 2

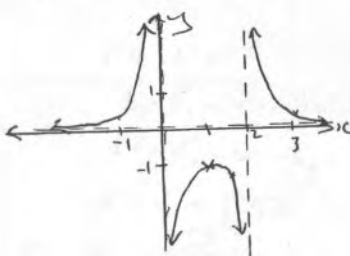
$$a) i) y = x^2 - 2x$$



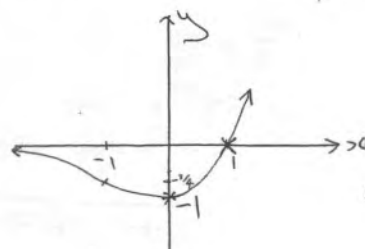
$$ii) y^2 = f(x)$$



$$iii) y = [f(x)]^{-1} = \frac{1}{x^2 - 2x}$$



$$iv) y = f(2^x) = 2^{2x} - 2^{x+1}$$



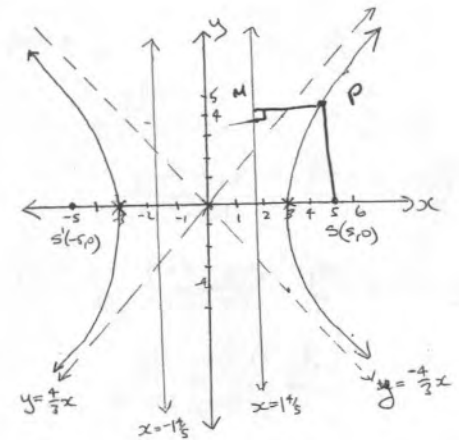
$$b) i) b^2 = a^2(e^2 - 1)$$

$$16 = 9(e^2 - 1) \quad (1)$$

$$\frac{16}{9} + 1 = e^2$$

$$e = \frac{5}{3} \quad (1)$$

ii)



(1) for directrices

(1) for vertices

(1) for asymptotes

(1) for focal shape

### Question 2

a)  $\frac{x^2}{100} + \frac{y^2}{75} = 1$

$a=10 \quad b=\sqrt{75}=5\sqrt{3}$

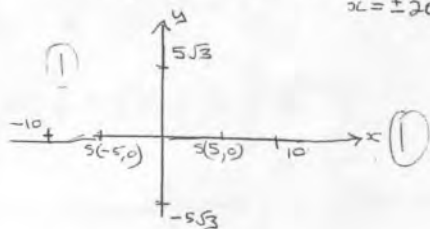
$b^2 = a^2(1-e^2)$

$\frac{75}{100} = 1-e^2$

$e^2 = \frac{1}{4}$

$e = \frac{1}{2}$

\* foci are  $(\pm 5, 0)$   
\* directrices are  $x = \pm 20$



ii) Differentiate with respect to  $x$ :

$\frac{2x}{100} + \frac{2y}{75} \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{x}{50} \cdot \frac{75}{2y} = -\frac{3x}{4y}$

① At  $P(5, 7\frac{1}{2})$   $\frac{dy}{dx} = \frac{-3(5)}{4(7\frac{1}{2})} = -\frac{1}{2}$

∴ gradient of normal is 2.

eqn. of normal is:

$y - 7\frac{1}{2} = 2(x - 5)$

$y = 2x - 2\frac{1}{2}$

$2y = 4x - 5$

ie  $x - 2y = 5$

iii) The intersection of the normals at P and Q will lie on the x-axis, the centre of the tangential circle is point of intersection of these 2 normals.

① sub. in  $y=0$  (lies on x-axis)

iv) radius  $= \sqrt{(5-1\frac{1}{4})^2 + (7\frac{1}{2}-0)^2}$   
 $= \sqrt{3\frac{3}{4}^2 + 7\frac{1}{2}^2}$   
 $= \sqrt{\frac{225}{16} + \frac{900}{16}}$   
 $= \sqrt{\frac{1125}{4}}$

eqn. of circle is:

$(x - 1\frac{1}{4})^2 + y^2 = \frac{1125}{16}$

$(4x - 5)^2 + 16y^2 = 1125$

b)  $\int_0^{2\pi} \frac{dx}{5 + 3\sin x + 4\cos x}$

$t = \tan \frac{x}{2}$  when  $x=0, t=0$   
 $x=2\pi, t=\sqrt{3}$

$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$

∴  $dx = 2\cos^2 \frac{x}{2} dt$

∴  $\int_0^{\sqrt{3}} \frac{\frac{2}{1+t^2} dt}{5 + \frac{6t}{1+t^2} + \frac{4-t^2}{1+t^2}} = \int_0^{\sqrt{3}} \frac{2 dt}{5 + 5t^2 + 4t - t^2}$

$= 2 \int_0^{\sqrt{3}} \frac{dt}{t^2 + 4t + 9} = 2 \int_0^{\sqrt{3}} \frac{dt}{(t+3)^2 + 2}$

①  $= -2 \left[ \frac{1}{t+3} \right]_0^{\sqrt{3}}$

$= \frac{-2}{\sqrt{3}+3} + \frac{2}{3}$

$= \frac{-2\sqrt{3}+6}{3-9} + \frac{2}{3}$

$= \frac{2\sqrt{3}-2}{6}$

$= \frac{\sqrt{3}-1}{3}$  Q.E.P.

c)  $\int \frac{dx}{x^2+x}$

$= \int \frac{1}{x} \frac{dx}{1+\frac{1}{x}}$

$= -\frac{1}{2} \int \frac{du}{u} = -\frac{1}{2} \ln(1+u) + C$

### Question 4

a)  $x^2 - y^2 = 9$

$2x - 2y \frac{dy}{dx} = 0$

∴  $\frac{dy}{dx} = \frac{x}{y}$

at  $P(p, q)$   $\frac{dy}{dx} = \frac{p}{q}$

eqn. of tangent at P is:

$y - q = \frac{p}{q}(x - p)$

$qy - q^2 = px - p^2$

$p^2 - q^2 = px - qy$

but P lies on the Hyperbola

∴  $p^2 - q^2 = 9$

∴ tangent is  $px - qy = 9$

Q.E.P.

ii) W lies on tangent at P and

the asymptote  $y = -x$

∴  $px + qx = 9$

$x = \frac{9}{p+q}$

∴ W is  $(\frac{9}{p+q}, \frac{-9}{p+q})$

Gradient of SW =  $\frac{0 - \frac{-9}{p+q}}{3e - \frac{9}{p+q}}$

$= \frac{9}{3e(p+q) - 9}$

$= \frac{3}{e(p+q) - 3}$

iii) T lies on  $y = x$  and on the tangent at P

∴  $px - qx = 9$

$x = \frac{9}{p-q}$

∴ T is  $(\frac{9}{p-q}, \frac{9}{p-q})$

Grad of ST =  $\frac{3}{3-e(p-q)}$

If  $\angle WST = \theta$  then

$\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2}$

$\tan \theta = \frac{\frac{3}{e(p+q)-3} - \frac{3}{3-e(p-q)}}{1 + \frac{9}{[e(p+q)-3][3-e(p-q)]}}$

①  $\tan \theta = \frac{3[3-e(p-q)] - 3[e(p+q)-3]}{[e(p+q)-3][3-e(p-q)] + 9}$   
 $= \frac{9 - 3ep + 3eq - 3ep - 3eq + 9}{(ep+eq-3)(3-ep+eq) + 9}$   
 $= \frac{18 - 6ep}{3ep - e^2p^2 + e^2pq + 3eq - e^2q^2 + 9}$

$= \frac{6(3-ep)}{6ep - e^2(p^2 - q^2)}$

①  $= \frac{6(3-ep)}{6ep - 9e^2} [as p^2 - q^2 = 9]$

$= \frac{6(3-ep)}{6ep - 18} [as e^2 = 2]$

$= \frac{6(3-ep)}{6(ep-3)}$   $p \neq \frac{3}{e} = \frac{3}{\sqrt{2}}$

∴  $\tan \theta = -1$

b)  $I_1$  is when  $n=1$

$I_1 = \int_0^{\pi/4} \tan x dx$

$= \int_0^{\pi/4} \frac{\sin x}{\cos x} dx$

$= [-\ln(\cos x)]_0^{\pi/4}$

$= -\ln \cos \frac{\pi}{4} + \ln \cos 0$

$= -\ln \frac{1}{\sqrt{2}} + \ln 1$

b) cont. (4/16)

ii) show  $I_n + I_{n-2} = \frac{1}{n-1}$

$$\begin{aligned} \text{L.H.S.} &= I_n + I_{n-2} \\ &= \int_0^{\pi/4} (\tan^n x) dx + \int_0^{\pi/4} \tan^{n-2} x dx \\ &= \int_0^{\pi/4} (\tan^n x + \tan^{n-2} x) dx \quad (1) \\ &= \int_0^{\pi/4} \tan^{n-2} x (\tan^2 x + 1) dx \\ &= \int_0^{\pi/4} \tan^{n-2} x \cdot \sec^2 x dx \quad (1) \\ &= \left[ \frac{1}{n-1} \tan^{n-1} x \right]_0^{\pi/4} \\ &= \frac{1}{n-1} [\tan^{n-1} \frac{\pi}{4} - \tan^{n-1} 0] \quad (1) \\ &= \frac{1}{n-1} (1 - 0) \\ &= \frac{1}{n-1} \\ &\quad Q.E.D. \end{aligned}$$

iii) let area represent  $I_n \therefore A_n = I_n$   
 now  $0 \leq x \leq \pi/4$   
 $\therefore 0 \leq \tan x \leq 1$

so  $0 \leq \tan^n x \leq \tan^2 x < \tan x \leq 1$  for  $n \geq 2$  (1)  
 i.e.  $0 \leq \tan^{n+2} x < \tan^n x < \tan^{n-2} x < \dots < \tan^2 x < \tan x \leq 1$  for  $n \geq 2$   
 now for  $\int_0^{\pi/4} \dots dx$

becomes

$$0 \leq I_{n+2} < I_n < I_{n-2} < \dots < I_2 < I_1 \leq \pi/4 \quad (1)$$

$$I_n < I_{n-2}$$

now  $I_{n+2} < I_n < I_{n-2}$   
 $I_{n+2} + I_n < 2I_n < I_n + I_{n-2}$

$$\frac{1}{n+1} < 2I_n < \frac{1}{n-1} \quad (1)$$

$$\frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)} \quad \text{for } n \geq 2$$

iv) when  $n=5 \Rightarrow \frac{1}{10+2} < I_5 < \frac{1}{10-2}$

$$\therefore \frac{1}{12} < I_5 < \frac{1}{8}$$

now  $I_5 + I_3 = \frac{1}{4}$  (from ii)

$$I_3 + I_1 = \frac{1}{2}$$

$$I_3 + \frac{1}{2} \ln 2 = \frac{1}{2} \quad (\text{as } I_1 = \frac{1}{2})$$

$$I_3 = \frac{1}{2} - \frac{1}{2} \ln 2 \quad (1)$$

$$\therefore I_5 = \frac{1}{4} - \left( \frac{1}{2} - \frac{1}{2} \ln 2 \right)$$

$$= -\frac{1}{4} + \frac{1}{2} \ln 2$$

$$\text{so } \frac{1}{12} < -\frac{1}{4} + \frac{1}{2} \ln 2 < \frac{1}{8} \quad (1)$$

$$\frac{1}{3} < \frac{1}{2} \ln 2 < \frac{3}{8}$$

$$\therefore \frac{2}{3} < \ln 2 < \frac{3}{4}$$

